GeneGuide AI: AI-Based Prenatal Genetic Risk Assessment and Decision Support System

Mohamed Ismail, Veerapuvin S, Sakthibalan G
Department of AI&ML, Vels Institute of Science, Technology & Advanced Studies (VISTAS), Pallavaram, Chennai, India
Project Guide: Ms. F. Benasir Begam

***Abstract*—GeneGuide AI is a web-based decision-support prototype that estimates prenatal genetic risk from parental details, family history, and pregnancy information. A random forest classifier, trained on a synthetic but medically informed dataset of 1,250 records with 14 input features, places each completed assessment into one of three categories: Low, Moderate, or High. The result is meant as an educational risk indicator, not a medical diagnosis. We built the system as a Flask application with user login, a validated multi-section assessment form, the prediction module, and a per-user history kept in SQLite. On a held-out test set of 250 records, the model reached 85.2% accuracy, with F1-scores of 0.92 for Low, 0.87 for Moderate, and 0.75 for High. The paper also reviews 12 related sources on machine-learning-based prenatal, genetic, and fetal risk assessment, clinical decision support, and family-history screening; places GeneGuide AI against them; describes the dataset, model, and implementation; and discusses the system's limitations and planned improvements.**

***Keywords*—Prenatal genetic risk, random forest, machine learning, decision support system, risk classification, Flask, synthetic dataset, explainable AI.**

I. INTRODUCTION

Prenatal care often begins with collecting family history, demographic, and screening information to determine whether a pregnancy carries a higher genetic risk. Today, a clinician usually makes this judgment by hand, weighing factors such as maternal age, consanguinity, carrier status, and earlier pregnancies against experience and published risk tables. This approach works, but it takes time, depends on the individual clinician's familiarity with the material, and is difficult to scale, as genetic counselors are few.

Machine learning can formalize part of this reasoning. A classifier trained on risk factors and outcomes provides a consistent and repeatable risk category for any given combination of inputs. This study presents GeneGuide AI, a student project that pairs such a classifier, a Random Forest trained on a synthetic, medically informed dataset of 14 prenatal and family-history features, with a complete web application, so that the prediction sits inside a tool that people can use from start to finish. We scope it clearly as an educational prototype. It relies on synthetic data rather than real patient records, and every output is presented as an AI-generated risk category to discuss with a qualified healthcare professional, never as a diagnosis.

This study makes the following contributions:

- A complete Flask web application covering registration, authenticated login, a validated fourteen-field assessment form, Random Forest prediction, a result page with a medical disclaimer, and a per-user history stored in SQLite.
- A random forest classifier trained and tested on a 1,250-record synthetic dataset, with precision, recall, F1-score, and the full confusion matrix reported.
- A literature review of 12 sources, covering polygenic-risk ensembling, first-trimester aneuploidy screening, Down syndrome prediction on imbalanced data, CTG-based fetal health classification, clinical decision-support outcomes, and family-history risk tools, which we use to pin down the gap GeneGuide AI addresses.
- A frank discussion of the prototype's limitations, along with a concrete plan for feature-importance explainability and comparison against other models.

II. RELATED WORK

A. Machine Learning in Clinical Decision Support

The integration of predictive models directly into clinical workflows has been a major research theme for approximately a decade. A 2023 scoping review of machine-learning-based clinical decision support found that most published systems are judged only on offline model accuracy. Far fewer studies have checked whether a deployed system changes clinicians' decisions, care delivery, or patient outcomes [9]. This gap between a model and a deployed system shapes this study. The Random Forest described in Section IV is just one part of a larger authenticated web application, and Section VI reports its evaluation, together with how the prediction is delivered to and used by a person, instead of treating the classifier as the whole contribution.

B. Ensembling Polygenic and Non-Genetic Risk Factors

Not every genetic risk study starts from family history questionnaires. Yuan et al. built DBNX, a pipeline that first merges 13 separately computed polygenic risk scores using an unsupervised deep belief network and then adds non-genetic variables such as age and lifestyle through XGBoost to give one composite risk score [1]. They tested it on four adult diseases using U.K. Biobank cohorts of tens of thousands of people. The DBN stage alone matched a supervised Super Learner baseline, and adding non-genetic factors beat polygenic scoring alone in every case. DBNX needs genome-wide genotyping data and biobank-scale cohorts, which a questionnaire-based prototype does not have. GeneGuide AI instead uses reported family and obstetric history in place of measured genetic variants, giving up fine-grained resolution in return for accessibility.

C. Neural-Network-Based First-Trimester Aneuploidy Screening

Neocleous, Nicolaides, and Schizas published two related studies that applied feed-forward artificial neural networks to first-trimester aneuploidy screening. Their inputs were clinically measured markers: maternal age, prior Trisomy-21 history, crown-rump length, PAPP-A, free β-hCG, and nuchal translucency, plus nasal bone visibility and Doppler flow in a longer configuration. The first study, trained on 51,208 pregnancies, reported a nine-marker network that identified every Trisomy-21 case in its evaluation set at a 3.9% false-positive rate. A two-stage variant that combined several binary networks caught 84.2% of other chromosomal abnormalities at a 6.4% false-positive rate [2]. The follow-up study returned to the problem with three larger datasets covering over 122,000 pregnancies. It compared raw marker values with multiples-of-median

normalization and tested whether shrinking the dominant euploid class through k-means clustering helped detect rarer aneuploidies. The best configuration again detected all Trisomy-21 cases, this time at a 4.8% false-positive rate, and caught 85.2% of other abnormalities [3]. Both studies relied on markers from supervised ultrasound scanning and laboratory assays, which a self-reported web form cannot reproduce. GeneGuide AI makes the opposite trade-off in this respect.

D. Cascaded and Ensemble Methods for Imbalanced Screening Data

Li et al. addressed a different obstacle: extreme class imbalance instead of raw accuracy [4]. Their maternal serum screening dataset from Jilin Province holds only 108 Down syndrome cases among more than 100,000 records, an imbalance of roughly 1:928 that defeats most classifiers. Their CVIFLR framework works in three stages. An isolation-forest anomaly pre-filter comes first, then a vote across several isolation forests trained on different partitions of the negative class, and finally a logistic-regression judgment on the much less skewed subset that remains. Compared with BalanceBagging, BalanceCascade, SMOTEENN, and SMOTETomek, it reached an AUROC of 0.99 and a 95% true-positive rate at a 4% false-positive rate. PAPP-A, β-hCG, and unconjugated estriol multiples-of-median, together with maternal age, emerged as the strongest features. Of the five base studies we reviewed, this one is closest in theme to GeneGuide AI, although the two differ significantly in terms of the severity of the imbalance and whether the inputs are lab-measured or self-reported.

E. Machine Learning for Cardiotocography-Based Fetal Monitoring

Salini et al. moved from prenatal screening to labor itself, classifying cardiotocography traces as normal, suspect, or pathological [5]. They compared six classifiers on 2,126 labeled records, and random forest performed best at 93% accuracy, ahead of logistic regression, decision tree, support vector classifier, voting classifier, and K-nearest neighbors. Their clinical question, whether a fetus is in distress right now, has little to do with GeneGuide AI's prenatal risk estimation. Still, the result is a useful data point. On a dataset close to ours in size and number of classes, random forest came out ahead of several alternatives, which offers some indirect support for the model choice in Section IV.

F. Family-History-Based Risk Assessment Tools

Family history is one of the oldest and cheapest tools for estimating genetic risk. A 2022 systematic review found 26 different family-history assessment tools already in use in primary care [11]. Its main conclusion is not flattering. Analytical and clinical validity were poorly reported for most tools, specificity was weak in about half, and detection of elevated-risk individuals ranged from 6.2% to 84.6% depending on the tool and condition. The reviewers argued that an ideal tool should be self-administered by patients instead of clinicians and should be connected with digital health records instead of remaining in paper form. GeneGuide AI follows this recommendation directly, as a self-administered, family-history-driven web questionnaire that produces an automated risk category. However, since it is an educational prototype trained on synthetic data, it should be read as a demonstration of the design pattern and not as a rival to the reviewed tools' validity benchmarks.

G. Comparative Analysis and Research Gap

TABLE I. COMPARISON OF RELATED WORK AND GENEGUIDE AI

Study	Data / Scale	Model	Focus
Yuan et al. [1]	UKBB, tens of thousands	DBN + XGBoost	Adult polygenic risk
Neocleous et al. [2]	51,208 cases	Feed-forward ANN	First-trimester T21/OCA
Neocleous et al. [3]	122,362 cases	ANN, k-means balancing	Raw vs. normalized, imbalance
Li et al. [4]	100,352 cases	Isolation forest+voting+LR	Down syndrome, extreme imbalance
Salini et al. [5]	2,126 CTG records	Random Forest	Intrapartum fetal health
Miroševič et al. [11]	26 tools, review	N/A (survey)	Family-history tool validity
GeneGuide AI	1,250 synthetic records	Random Forest	Family-history prenatal risk, web app

Taken together, the seven studies in Table I approach pregnancy-related and family risk from several independent directions. They cover polygenic-risk ensembling on large genotyped biobank cohorts [1], first-trimester screening based on ultrasound and biochemical measurements [2], [3], serum-marker and cardiotocography classification, one under severe class imbalance [4], [5], the wider question of what makes a clinical decision-support system change outcomes instead of only performing well offline [9], and a critical audit of family-history tools already used in primary care, few of which are rigorously validated or self-administered [11]. None of them combines a family-history-driven, three-tier risk indicator with the self-administered, records-adjacent delivery that the family-history literature itself recommends. GeneGuide AI does not claim the diagnostic precision of the large, clinically validated systems discussed in Sections II-B through II-E. Its contribution is narrower and more practical: a reproducible demonstration, on a synthetic dataset, of how a Random Forest classifier can be delivered end to end as the kind of self-administered, authenticated web tool that both the decision-support literature [9] and the family-history literature [11] describe as underused.

III. PROPOSED SYSTEM

A. System Overview

GeneGuide AI is a Flask web application. It collects 14 prenatal, family history, and screening-related inputs through a structured form, passes them to a trained random forest classifier, and returns one of three risk categories (low, moderate, or high) with plain-language guidance. It does not attempt to diagnose any specific disorder. Each result page carries a disclaimer that describes the output as an educational risk category and not a medical conclusion, and it encourages the user to see a qualified healthcare professional.

B. Input Factors

TABLE II. INPUT FEATURES BY CATEGORY

Category	Features
Demographic / Pregnancy	Mother_Age, Father_Age, Pregnancy_Week
Family History	Family_History, Affected_Members, Type_of_Family_History, Carrier_Status
Genetic Testing & Obstetric History	Genetic_Test_Result, Prev_Child_Disorder, Prev_Abnormal_Pregnancy, Consanguineous_Marriage,

	Prev_Miscarriages, Multiple_Pregnancy, Prev_Prenatal_Screening

C. System Architecture

The application uses a conventional three-tier web architecture. The presentation tier is made of Jinja2-rendered HTML templates styled with CSS, plus a little client-side JavaScript that checks the form for consistency. The application tier is a Flask server with routes for registration, login, the assessment form, prediction, result display, and history retrieval. At startup, it loads the trained Random Forest model, its label encoders, and its target encoder using joblib. The data tier is a SQLite database, geneguide.db, with a users table and an assessments table joined by a foreign key; thus, each stored assessment belongs to exactly one authenticated user.

D. Workflow

A new user registers with a username and password, and the password is stored only as a Werkzeug-generated hash. After login, a Flask session records the user, and the login_required decorator limits the assessment and history routes to authenticated sessions. When the user submits the assessment form, the server checks the numeric ranges, encodes the categorical fields with the same label encoders used in training, obtains a prediction from the Random Forest, saves the inputs and predicted risk level in the assessments table, and renders the result page. Users can look through all of their own past assessments on a history page and can log out at any time, which clears the session.

IV. METHODOLOGY

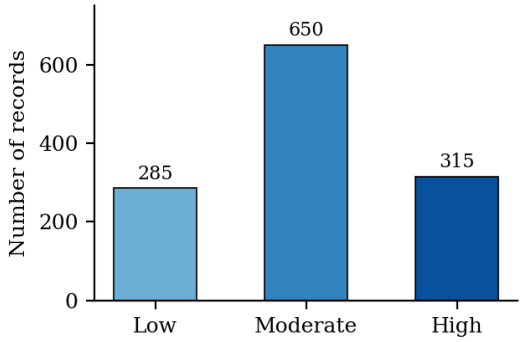
A. Dataset Preparation

The model was trained on a synthetic, medically informed dataset of 1,250 records. Each record has 14 input features and one target column, Risk_Level, with the values Low, Moderate, or High. Using synthetic data instead of real patient records for prototype development is an accepted practice in healthcare machine learning, largely because it avoids the privacy, consent, and data-sharing limits that restrict access to clinical records [10]. The dataset is meant for prototype and educational evaluation only, and none of the reported results should be taken as evidence of real pregnancies. The class distribution leans moderately toward the Moderate category: 650 records (52.0%) are Moderate, 315 (25.2%) are High, and 285 (22.8%) are Low.

TABLE III. CLASS DISTRIBUTION

Risk_Level	Records	Share
Low	285	22.8%
Moderate	650	52.0%
High	315	25.2%

Fig. 1. Class Distribution of the Training Dataset



B. Data Preprocessing

Numeric fields (maternal and paternal age, pregnancy week, number of affected family members, and prior miscarriages) go into the model as they are, and each is also bounded separately by the web form's input validation. One preprocessing detail turned out to matter during development. Several categorical fields in the dataset use the literal text value "None" to mean that a condition is absent, and common data-loading libraries can silently read this text as a missing value instead of a valid category. If nobody corrects it, this can quietly distort the relationship the model learns between those fields and the target, for every record where the category should have counted as informative. We therefore keep such placeholder categories explicitly and do not rely on default missing-value handling. It is worth documenting for anyone who extends this dataset or works with a similarly structured one.

C. Feature Encoding

Nine of the 14 input features were categorical: Family_History, Type_of_Family_History, Carrier_Status, Parental_Genetic_Test_Result, Previous_Child_Genetic_Disorder, Previous_Abnormal_Pregnancy, Consanguineous_Marriage, Multiple_Pregnancy, and Previous_Prenatal_Screening. Each was converted with its own scikit-learn LabelEncoder [7] fitted on the training data. The same encoders were saved with joblib and reused at prediction time, so a category maps to the same integer in training and in live use. The target column has a separate LabelEncoder of its own, and predictions are inverse-transformed back to a Low, Moderate, or High label.

D. Random Forest Model

A RandomForestClassifier [7] is the only classification model in this version of the system. It uses 200 estimators and a fixed random state of 42 for reproducibility and follows the ensemble-of-decision-trees approach that Breiman first described [6]. We chose Random Forest because it copes well with tabular data that mixes categorical and numeric features, it resists overfitting better than a single decision tree, and it can later expose feature importances for explainability. The CTG classification study in Section II-E also found the algorithm a good fit for a multi-class dataset of similar size [5].

E. Train-Test Procedure

The dataset is divided 80/20 into 1,000 training records and 250 test records, using a fixed random state of 42 and stratifying on the target column so that both splits maintain the Low, Moderate, and High proportions. The model is fitted on the training split only, and every metric in Section VI is computed on the held-out 250-record test split.

F. Risk Classification

At inference time, the fourteen validated form inputs are placed into a single-row DataFrame, the categorical columns are transformed using the saved encoders, and the trained Random Forest predicts a class, which is inverse-transformed to Low, Moderate, or High. The result page shows no numeric confidence score. We deliberately limit the output to the categorical label and its educational guidance.

V. IMPLEMENTATION

A. Web Application

The front end has six Jinja2 templates: a home page that describes the project and its purpose, an assessment page, a result page, and login, registration, and history pages. The assessment page holds a fourteen-field form, split into basic information,

family history, and genetic and pregnancy information. A single CSS file handles styling. A small JavaScript file keeps the family-history radio buttons in sync with the type-of-family-history dropdown, so that a user who reports no family history is steered to the matching "None" category and does not end up with an inconsistent combination.

B. Backend

The backend is a single-file Flask application. Registration hashes the submitted password with Werkzeug's generate_password_hash before inserting it into SQLite, and login checks it with check_password_hash against the stored hash; thus, plaintext passwords are never compared or stored. All SQL statements use parameterized queries, which avoid SQL injection risk. A login_required decorator on the assessment, prediction, and history routes sends unauthenticated requests to the login page.

C. Database

The application uses one SQLite database file, geneguide.db, which is created automatically on first run if it does not exist. The users table stores an auto-incrementing id, a unique username, and a hashed password. The assessments table stores an auto-incrementing id, a user_id foreign key, a timestamp, the fourteen assessment inputs, and the predicted risk_level. Every read from the assessments table on the history page is filtered by the logged-in user's user_id; therefore, one user's history is never visible to another.

D. Prediction Module

The /predict route starts by checking that numeric fields fall within clinically plausible ranges: maternal age 18 to 45, paternal age 20 to 55, pregnancy week 8 to 40, affected family members 0 to 3, and prior miscarriages 0 to 3. If a field is missing or out of range, the route returns a descriptive error message, so that the model never receives unvalidated input. Once the inputs pass, the module builds the feature DataFrame, applies the saved label encoders, obtains a prediction from the Random Forest, stores the assessment and its predicted risk level in the database, and renders the result template with the Low, Moderate, or High label.

VI. RESULTS AND DISCUSSION

A. Dataset Distribution

Table III shows that the Moderate class makes up just over half of all records, while High and Low each account for roughly a quarter. Fig. 1 shows the same split visually. This imbalance is moderate and much milder than those handled in [3] and [4], but it still affects per-class performance, especially the lower recall for the High-risk class discussed below.

B. Model Performance

On the 250-record held-out test split, the Random Forest model achieved an overall accuracy of 85.2%.

C. Classification Report

TABLE IV. PER-CLASS CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
Low	0.93	0.91	0.92	57
Moderate	0.82	0.92	0.87	130
High	0.86	0.67	0.75	63

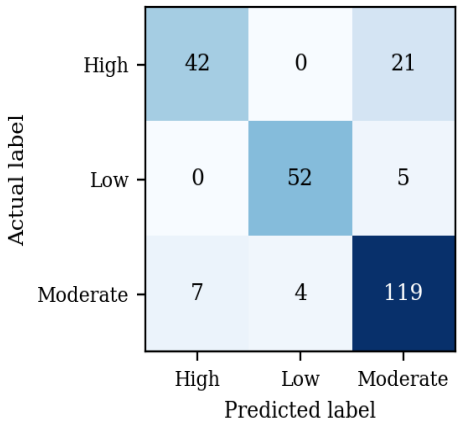
Low-risk cases show the best balance of precision and recall (F1 = 0.92). Moderate-risk cases, the majority class, are recalled most often (0.92) but with somewhat lower precision (0.82), which means some Low- and High-risk cases are being pulled into the Moderate prediction. High-risk cases have the lowest recall (0.67) of the three classes, despite a fairly high precision (0.86). In other words, when the model predicts high risk, it is usually correct, but it misses a meaningful share of the real high-risk cases.

D. Confusion Matrix

TABLE V. CONFUSION MATRIX (ROWS = ACTUAL, COLUMNS = PREDICTED)

Actual \ Predicted	High	Low	Moderate
High	42	0	21
Low	0	52	5
Moderate	7	4	119

Fig. 2. Confusion Matrix on the Held-Out Test Set



The confusion matrix shows that the model never mixed up High and Low risk in either direction. The zero entries in the High/Low and Low/High cells mean that when the model is wrong, it lands in the neighboring Moderate category and not at the opposite end of the scale, which is a reassuring failure pattern for a decision-support tool. The largest single source of error is the 21 truly High-risk cases predicted as Moderate, and this accounts for most of the shortfall in High-risk recall noted above. A smaller pattern is also visible: 7 Moderate cases were predicted as High and 4 as Low.

E. Discussion

These results are in accordance with what we would expect from a Random Forest trained on a moderately imbalanced three-class synthetic dataset of this size. The majority Moderate class is recalled most reliably, the two minority classes trade precision against recall in different ways, and errors cluster in the neighboring class instead of spanning the whole risk range. Compared with the much larger clinically collected datasets in [2]–[4], the 1,250 records here put a natural ceiling on how finely the boundary between High and Moderate risk can be learned, and that is the most plausible explanation for the lower High-risk recall. The absence of any High/Low or Low/High confusion is a good property for an educational tool, because the most serious kind of error, mistaking a truly high-risk case for one needing no extra attention, does not occur in the test set. High-risk cases that are missed are still flagged as Moderate and not Low, which builds in a degree of caution. The strong showing of Random Forest also agrees with the CTG-based study in Section II-E, where the same algorithm was the best of six candidates on a multi-class dataset of similar size [5].

VII. LIMITATIONS AND FUTURE WORK

GeneGuide AI has several limitations that readers should consider before drawing conclusions beyond its educational scope.

- The training data were synthetic and not drawn from real patient records; therefore, the relationships the model learned between risk factors and outcome reflected how the dataset was constructed and not verified clinical epidemiology.
- At 1,250 records, the dataset is small compared to the tens of thousands of cases used in the clinically validated systems reviewed in Section II. This limits how finely the model can separate neighboring risk categories, as shown by the high-risk recall. The moderate imbalance toward the moderate class (Table III) could be handled in a later revision with resampling, such as SMOTE [8], which several of the reviewed studies used for much more severe imbalances than ours.
- The system has had no clinical validation. Its accuracy figures describe the performance on a held-out split of the same synthetic dataset, not agreement with real diagnostic outcomes.
- The model performance depends entirely on how well the training data cover realistic input combinations. A case far outside the dataset's distribution may result in an unreliable prediction with no explicit warning.
- The system should not replace consultation with a qualified healthcare professional, and the medical disclaimer on every result page states so.

Future work includes computing and displaying Random Forest feature importances and evaluating post-hoc explanation methods such as LIME and SHAP, which have been shown to make Random Forest predictions on tabular healthcare data much easier for non-specialists to interpret [12]. This would allow each result to come with the inputs that influenced it the most. We also plan to compare Random Forest with Logistic Regression and Decision Tree classifiers on the same dataset, choosing the model from documented evaluation metrics and not by assumption, and to add k-fold cross-validation for a more stable estimate of generalization performance. Other planned steps include expanding the synthetic dataset with more records and a more deliberately balanced class distribution and adding a simple dashboard that summarizes a user's own assessment history, including counts by risk level over time.

VIII. CONCLUSION

This paper presented GeneGuide AI, a Flask-based prototype for prenatal genetic risk assessment and decision support, driven by a Random Forest classifier and set against a twelve-source literature review. The review covered polygenic-risk ensembling, first-trimester aneuploidy screening, Down syndrome prediction on imbalanced data, cardiotocography-based fetal health classification, clinical decision-support outcomes and the validity of family-history screening tools. On a held-out test set of 250 synthetic records the model reaches 85.2% accuracy, and its errors tend to fall in the adjacent category and not in the more serious High/Low mix-ups. The system is deliberately scoped as an educational prototype: it uses a synthetic dataset, has not been clinically validated, and presents its output strictly as an AI-generated risk category and not a diagnosis, pointing users to professional medical advice for real decisions. Planned work on explainability, model comparison and a larger, more carefully built dataset should strengthen its value as a demonstration of applied machine learning in decision support, while keeping the educational framing throughout.

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